

Phase 1B.5 — Opportunity Portfolio Optimization

Patent-Oriented Embedded Systems Innovation Pipeline

Board: CTO · IEEE Embedded Systems Researcher · Patent Examiner · VC Partner · Industrial Product Architect · Embedded AI Engineer · Cybersecurity Researcher · Systems Engineer · Innovation Strategist · Engineering Design Review Committee **Source of truth:** Phase 1B Opportunity Analysis (32 clusters, structured records, 11-criterion scores) — no new research performed, no clusters re-derived **Objective:** Apply hard-constraint gates to the 32 Phase 1B clusters and decide which deserve to advance into Phase 1C

Method Note

Phase 1B produced eleven soft-weighted scores per cluster and a composite ranking. That composite answers *"how good is this opportunity in general?"* Phase 1B.5 asks a different, harder question: *"can **this specific team, with this specific budget, in this specific timeline**, actually execute this?"* — which is why a cluster can rank highly in Phase 1B and still fail a gate here, and vice versa.

Gates 1, 2, 3, and 6 are **critical (PASS/FAIL)** — failure normally eliminates a cluster. Gates 4, 5, 7, and 8 are **informational (scored)** — they classify quality among survivors but do not themselves eliminate anything, per the task specification.

Two scoring notes carried over directly from Phase 1B's own eleven-criterion data (to avoid re-deriving anything):

- Gate 1 (Semester Feasibility) draws on Phase 1B's "Semester Feasibility" score and its own explicit prototype-scoping language.
- Gate 2 (Budget) draws on Phase 1B's "Budget Feasibility" score and cost estimate.
- Gate 3 (Embedded Systems Necessity) draws on Phase 1B's "Embedded Systems Advantage" score.
- Gate 4 (Patent/Research White Space) draws on Phase 1B's Patent Potential + Research Gap scores.

Gates 6, 5, 7, and 8 required fresh judgment against the specific team profile (3 CS/Cybersecurity engineering students; strong in Embedded C/C++, STM32, ESP32, Raspberry Pi, Linux, networking, cybersecurity, IoT, AI-assisted development, hardware integration; willing to learn adjacent skills) — this is genuinely new evaluation, not a repeat of Phase 1B.

A key pattern surfaced repeatedly during Gate 1/2/6 review: several high-value clusters "fail" only because Phase 1B's own record already flagged that the *general/full* version of the

opportunity is unrealistic in two months, while a *narrowly-scoped* version is explicitly called realistic in that same record (e.g., "a single-leak proof-of-concept is more realistic," "a narrowly-scoped single-protocol demo is realistic"). Where this pattern held, and the cluster showed no compensating skill/budget/necessity problem, the board treated it as **exceptional justification** rather than an automatic kill — the cluster survives, but only under an explicit, mandatory scope restriction noted alongside it. Where a cluster failed a gate *and* lacked this narrow-scope escape hatch (e.g., it required specialized domain expertise — biosensor chemistry, advanced applied cryptography research, dedicated ML research — outside what "willing to learn adjacent skills" can plausibly cover in eight weeks), no exception was granted.

1. Gate Evaluation Table (All 32 Clusters)

Legend: G1=Semester Feasibility · G2=Budget · G3=Embedded Necessity · G4=Patent/White Space (H/M/L) · G5=Demo Impact (1-10) · G6=Technical Capability · G7=Learning Return (1-10) · G8=Innovation Density (1-10). P*=Pass with mandatory scope condition noted.

ID	Cluster (short name)	Dom	G1	G2	G3	G4	G5	G6	G7	G8	Ver
OC-01	Offline RPM / Vital-Sign Monitoring	H	PASS	PASS	PASS	MED	6	P* (use off-shelf AFE)	7	6	SUI
OC-02	Non-Contact Fall Detection	H	P*	P*	PASS	HIGH	8	PASS	8	8	SUI
OC-03	Point-of-Care Diagnostics	H	PASS	PASS	PASS	MED	7	PASS	7	6	SUI
OC-04	Medication Adherence	H	PASS	PASS	FAIL	LOW	4	PASS	5	3	ELI
OC-05	Cold-Chain / Pharma Logistics	H	PASS	PASS	PASS	HIGH	6	PASS	7	7	SUI
OC-06	Hospital Asset Tracking + Security	H	PASS	PASS	PASS	MED	6	PASS	8	7	SUI
OC-07	Decoupled Continuous Glucose Monitor	H	FAIL	FAIL	PASS	HIGH	6	FAIL	6	6	ELI
OC-08	Assistive Communication & Mobility	A	P*	P*	PASS	HIGH	9	P*	8	8	SUI
OC-09	Navigation/Obstacle Detection (VI)	A	PASS	PASS	PASS	LOW	7	PASS	6	5	SUI
OC-10	Vibration-Based Predictive Maintenance	I	PASS	PASS	PASS	MED	7	PASS	8	7	SUI
OC-11	Secure Offline IIoT/OT Anomaly Detection	I	P*	PASS	PASS	HIGH	6	PASS	9	8	SUI
OC-12	Industrial Process Quality Monitoring	I	PASS	PASS	PASS	HIGH	6	P* (needs shop access)	7	6	SUI

ID	Cluster (short name)	Dom	G1	G2	G3	G4	G5	G6	G7	G8	Ver
OC-13	Worker Safety Monitoring	I	PASS	PASS	PASS	MED	6	PASS	7	6	SUI
OC-14	Energy-Infrastructure Monitoring	I	P*	P*	PASS	HIGH	6	P*	7	7	SUI
OC-15	Precision Irrigation & Soil Sensing	E	PASS	PASS	PASS	MED	7	PASS	7	6	SUI
OC-16	Pest/Crop-Disease/Livestock Monitoring	E	PASS	PASS	PASS	MED	6	PASS	7	6	SUI
OC-17	Self-Calibrating Air/Water Quality	E	PASS	PASS	PASS	MED	6	PASS	7	5	SUI
OC-18	Water-Pipeline Leak Detection (DSP)	E	FAIL	PASS	PASS	HIGH	7	FAIL	8	8	ELI
OC-19	Climate & Disaster Early-Warning	E	P*	PASS	PASS	MED	6	PASS	7	6	SUI
OC-20	Wildlife & Biodiversity Monitoring	E	PASS	PASS	PASS	LOW	5	PASS	6	5	SUI
OC-21	Hardware Root-of-Trust & Secure Boot	C	PASS	PASS	PASS	HIGH	7	PASS	9	7	SUI
OC-22	Side-Channel Attack Defense (PUF/EM)	C	P*	P*	PASS	HIGH	8	P* (equipment access)	9	9	SUI
OC-23	Post-Quantum Cryptography	C	FAIL	P*	PASS	HIGH	5	FAIL	9	7	ELI
OC-24	Smart Parking & Occupancy Sensing	S	PASS	PASS	FAIL	LOW	4	PASS	4	3	ELI
OC-25	Pothole/Road Condition	S	PASS	PASS	PASS	MED	7	PASS	7	6	SUI

ID	Cluster (short name)	Dom	G1	G2	G3	G4	G5	G6	G7	G8	Ver
Assessment											
OC-26	Structural Health Monitoring	S	PASS	PASS	PASS	MED	6	PASS	7	6	SU
OC-27	HVAC Occupancy / Indoor Air Quality	S	PASS	PASS	FAIL	LOW	4	PASS	5	4	EL
OC-28	Asset/Cargo/Tool Anti-Theft Tracking	L	PASS	PASS	PASS	MED	6	PASS	7	6	SU
OC-29	Retail Inventory Automation / ESL	L	PASS	PASS	FAIL	LOW	4	PASS	4	3	EL
OC-30	Battery-Free Energy-Harvesting Nodes	X	FAIL	P*	PASS	HIGH	6	P*	8	7	EL
OC-31	Edge-AI Research Frontier	X	FAIL	P*	PASS	HIGH	6	FAIL	9	9	EL
OC-32	Consumer Smart-Home Safety Sensors	X	PASS	PASS	FAIL	LOW	5	PASS	5	4	EL

Result: 22 of 32 clusters survive (69%) — 16 with a clean pass on all four critical gates, 6 with a conditional pass requiring a mandatory, explicit scope restriction (marked SURVIVE* and detailed in Section 3). 10 clusters are eliminated.

(This 69% survival rate is higher than the "typical 40-60%" the task anticipates. Section 3's closing note addresses this directly rather than silently forcing a lower number.)

2. Eliminated Opportunities — Reasons

Every elimination below is a gate failure with **no compensating exceptional circumstance** — either the failure was compounded across multiple gates, or no narrow-scope path existed to rescue it (unlike the six SURVIVE* clusters in Section 1, where a narrow-scope path did exist and was used to justify survival).

OC-04 — Medication Adherence & In-Home Care Monitoring

Failed Gate 3 (Embedded Necessity). Phase 1B's own record notes "standalone operation is useful but not indispensable" — a smartphone-app-based reminder system with a Bluetooth pill-bottle sensor would solve the actual problem nearly as well. With the lowest Patent Potential (3) and Research Gap (4) scores of any healthcare cluster, there's no compensating novelty to justify keeping a project where embedded hardware isn't the load-bearing element. **Eliminated — not because it's a bad idea, but because it isn't an embedded-systems idea.**

OC-07 — Decoupled Continuous Glucose / Metabolic Monitoring

Failed Gate 1, Gate 2, and Gate 6 simultaneously. The compounding is the problem: Semester Feasibility was already marginal (5/10), Budget Feasibility was under pressure from sensor front-end sourcing (5/10), and — most decisively — the core technical challenge is glucose-sensing *chemistry/biosensor engineering*, a domain outside even an ambitious "willing to learn" stretch for a CS/Cybersecurity team in eight weeks. Phase 1B itself flagged the underlying sensing technology as "patent-encumbered." Unlike OC-11 or OC-22, there is no narrow re-scoping that removes the biosensor-chemistry dependency — even the decoupled-architecture-only version still needs a working glucose sensor to demonstrate anything meaningful. **Eliminated — genuine domain-expertise mismatch, not merely a scoping problem.**

OC-18 — Water-Pipeline Leak Detection (Acoustic/RF DSP)

Failed Gate 1 and Gate 6. Phase 1B explicitly flagged this as "High difficulty" and noted that "a full multi-leak localization demo is unrealistic in 2 months." The board considered whether a narrow single-leak demo (the same rescue pattern used for OC-11/OC-22) could save it, but the underlying technical bottleneck here is different in kind: it requires genuinely advanced blind-source-separation/beamforming DSP research (ICA, MUSIC-like algorithms) that sits outside the team's stated strengths (embedded, networking, cybersecurity, AI-assisted development) and isn't something a security-focused team would plausibly pick up competently in the timeframe, even narrowly scoped. Environment/agriculture is already well-represented among survivors (OC-15, OC-16, OC-17, OC-19, OC-20), so no diversity argument compensates. **Eliminated.**

OC-23 — Post-Quantum Cryptography for Resource-Constrained Devices

Failed Gate 1 and Gate 6. This is the single highest research-gap/novelty cluster in the whole Phase 1B dataset, and the board debated it hardest. But Phase 1B's own language is unusually direct here: "genuinely challenging to build a credible, working demo in 2 months without prior cryptography-engineering background," and appropriate only "if the team has unusually strong applied-cryptography background" — which is a materially deeper and narrower specialization than general "cybersecurity," and nothing in the team profile

confirms it. Unlike OC-11/OC-22 (where the team's general embedded-security skillset directly transfers), PQC requires dedicated lattice-cryptography and side-channel-aware implementation expertise that isn't a scoping problem — it's a background problem.

Eliminated from this semester's portfolio; flagged as a strong candidate for a future extension or thesis-level follow-on if a team member develops the specific cryptography background.

OC-24 — Smart Parking & Urban Occupancy Sensing

Failed Gate 3 (Embedded Necessity). "Useful but not indispensable — cameras/other approaches also viable," per Phase 1B. Combined with the lowest Patent Potential (3), Research Gap (3), Learning Return (4), and Innovation Density (3) scores of any smart-city cluster, this is the cleanest possible elimination case: easy, cheap, and technically unremarkable. **Eliminated.**

OC-27 — Building Occupancy-Driven HVAC/Lighting & Indoor Air Quality

Failed Gate 3 (Embedded Necessity). Same pattern as OC-24 — "local privacy-preserving inference is useful but not strictly indispensable." An established, commoditized building-automation-optimization space with limited technical differentiation (Patent 4, Research Gap 4). **Eliminated.**

OC-29 — Retail Inventory Automation & Electronic Shelf Labels

Failed Gate 3 (Embedded Necessity). "The core value is cost/openness, not a unique embedded capability," per Phase 1B. This cluster also carried the lowest composite score (5.7) of all 32 clusters in Phase 1B, and nothing in the gate review changes that assessment — it is simply the weakest opportunity in the dataset on every axis that matters for a patent-oriented project. **Eliminated.**

OC-32 — Consumer Smart-Home Predictive Safety Sensors

Failed Gate 3 (Embedded Necessity). "Local fusion is useful but the category is not fundamentally embedded-dependent," per Phase 1B. Crowded, brand-dominated consumer market with limited patentable differentiation (Patent 4, Research Gap 4). **Eliminated.**

OC-30 — Energy-Autonomous / Battery-Free Wireless Sensor Node Platforms

Failed Gate 1, with a Gate 6 caveat (multi-source energy-harvesting circuit design is specialized analog power-electronics work, adjacent to but not squarely within the team's stated strengths). Phase 1B itself already suggested the more useful framing: this may be "more strategically valuable as a *feature* integrated into another cluster's prototype... than as a standalone semester project." The board agrees — **eliminated as a standalone cluster**, but explicitly recommended as an optional enhancement to whichever surviving cluster is ultimately chosen in Phase 1C (most naturally OC-15, OC-19, OC-20, or OC-28, all of which

involve remote/distributed sensor nodes that would benefit from reduced battery-maintenance burden).

OC-31 — Edge-AI Research Frontier (federated learning, neuromorphic vision, neurosymbolic AutoML)

Failed Gate 1 and Gate 6. This cluster has the single highest Research Gap (10) and AI Integration (10) scores in the entire dataset — genuinely the most exciting research territory here. But Phase 1B's own conclusion is unambiguous: it is "appropriate primarily for a team explicitly prioritizing publication/patent outcomes over commercial speed," which conflicts directly with this team's stated multi-goal profile (patent **and** research publication **and** excellent semester project **and** startup foundation, with a hard 2-month/ ₹3-5k constraint). This is a case where the opportunity is excellent but the *fit to this specific team and constraint set* is poor. **Eliminated from the semester portfolio; flagged as the strongest candidate in the entire dataset for a post-semester research extension or a fourth team member with dedicated ML-research background.**

3. Survivor List (22 Clusters)

3.1 Clean Survivors — No Scope Conditions (16 clusters)

ID	Cluster	Domain	Phase 1B Composite	G4	G7/ G8
OC-01	Offline RPM / Vital-Sign Monitoring	H	7.5	MED	7 / 6
OC-03	Point-of-Care Diagnostics	H	7.3	MED	7 / 6
OC-05	Cold-Chain / Pharma Logistics Monitoring	H	7.4	HIGH	7 / 7
OC-06	Hospital Asset Tracking + Security	H	6.9	MED	8 / 7
OC-09	Navigation/Obstacle Detection (VI)	A	6.7	LOW	6 / 5
OC-10	Vibration-Based Predictive Maintenance	I	8.1	MED	8 / 7
OC-12	Industrial Process Quality Monitoring	I	7.4	HIGH	7 / 6

ID	Cluster	Domain	Phase 1B Composite	G4	G7/ G8
OC-13	Worker Safety Monitoring	I	7.0	MED	7 / 6
OC-15	Precision Irrigation & Soil Sensing	E	7.9	MED	7 / 6
OC-16	Pest/Crop-Disease/Livestock Monitoring	E	6.9	MED	7 / 6
OC-17	Self-Calibrating Air/Water Quality	E	7.2	MED	7 / 5
OC-20	Wildlife & Biodiversity Monitoring	E	6.2	LOW	6 / 5
OC-21	Hardware Root-of-Trust & Secure Boot	C	8.3	HIGH	9 / 7
OC-25	Pothole/Road Condition Assessment	S	6.4	MED	7 / 6
OC-26	Structural Health Monitoring	S	7.5	MED	7 / 6
OC-28	Asset/Cargo/Tool Anti-Theft Tracking	L	6.6	MED	7 / 6

(OC-12 carries one logistics caveat, not a gate failure: realistic validation needs physical access to a welding shop or CNC machine, which the team should confirm before committing.)

3.2 Conditional Survivors — Mandatory Scope Restriction (6 clusters)

These passed Gate 3 and Gate 6 (core team-skill fit is genuine) but only clear Gate 1 and/or Gate 2 if the team commits *now*, at the Phase 1C idea-synthesis stage, to the narrower version of the opportunity described. This is not a soft suggestion — Phase 1C should not re-open the broader version of any of these.

ID	Cluster	Domain	Composite	Mandatory Scope Restriction
OC-02	Non-Contact Elderly Fall Detection	H	7.6	Use IMU + acoustic sensing as the primary modality; treat mmWave radar as an optional stretch add-on only if budget allows, not the core design
OC-08	Assistive Communication & Mobility Devices	A	8.0	Scope to the Braille-display or AAC/sign-language-translation sub-opportunity; explicitly exclude the EMG-prosthetic-hand sub-opportunity, which requires biomedical-signal and mechanical-actuator expertise outside the team's profile
OC-11	Secure Offline IIoT/OT Anomaly Detection	I	8.2	Scope to one legacy protocol (e.g., Modbus) and one anomaly-detection demo; do not attempt a general-purpose multi-protocol gateway
OC-14	Energy-Infrastructure Monitoring	I	7.7	Scope to solar-panel micro-crack monitoring or battery state-of-health; explicitly exclude the live-power-line TDR anti-theft sub-opportunity, which carries both an electrical-safety risk and a specialized-skill gap
OC-19	Climate & Disaster Early-Warning Networks	E	7.3	Scope to a single hazard type and a single node (e.g., flood water-level sensor); do not attempt multi-hazard precursor validation
OC-22	Physical & Side-Channel Attack Defense	C	7.8	Scope to one narrow, well-bounded demonstration (e.g., PUF key generation from SRAM power-up state on an ESP32); confirm access to basic measurement equipment (oscilloscope or logic analyzer) before committing

4. Portfolio Diversity Analysis

Domain Distribution of the 22 Survivors

Domain	Survivors	Count	% of Portfolio
Healthcare (H)	OC-01, 03, 05, 06	4	18%
Assistive Technology (A)	OC-08, 09	2	9%
Industrial (I)	OC-10, 11, 12, 13, 14	5	23%
Agriculture/Environment (E)	OC-15, 16, 17, 19, 20	5	23%
Cybersecurity-Hardware (C)	OC-21, 22	2	9%
Smart City/Infrastructure (S)	OC-25, 26	2	9%
Supply Chain/Logistics (L)	OC-28	1	4.5%
Cross-Cutting Enabling Tech (X)	(none)	0	0%

Is the Portfolio Over-Concentrated?

No single domain dominates. Industrial and Agriculture/Environment tie at 23% each, Healthcare sits at 18% — no domain exceeds roughly one-quarter of the surviving portfolio. This is a meaningfully healthier distribution than a naive reading of the raw Phase 1A source material would suggest (recall the four Deep Research reports individually skewed heavily toward healthcare and assistive-tech narratives, especially Qwen's report) — the gate system's Embedded-Necessity filter (Gate 3) did real work here, disproportionately removing weaker healthcare/consumer clusters (OC-04, OC-07) and smart-city/consumer clusters (OC-24, OC-27, OC-29, OC-32) rather than industrial or environmental ones, which naturally rebalanced the portfolio toward domains where embedded hardware is more clearly load-bearing.

Three domains are thin but not necessarily under-represented: Assistive Technology, Cybersecurity-Hardware, and Smart City/Infrastructure each carry only 2 survivors. This looks concerning only if it's read as a raw count. Read qualitatively, it isn't:

- **Cybersecurity-Hardware (OC-21, OC-22)** — only 2 clusters, but they are ranked **#1 and #6 overall** in Phase 1B's composite ranking, carry the two highest Patent Potential-adjacent (HIGH/HIGH) classifications, and are the *most direct match* to the team's stated Computer Science/Cybersecurity specialization. This is a case of quality

concentration, not weakness — it would be a mistake for Phase 1C to "dilute" this domain in the name of diversity when it's arguably the team's strongest strategic fit.

- **Assistive Technology (OC-08, OC-09)** — thin in count, but OC-08 alone carries the single highest Patent Potential score (9) in the entire 32-cluster dataset (the Braille-actuator sub-opportunity specifically).
- **Smart City/Infrastructure (OC-25, OC-26)** — genuinely the weaker end of the thin group; neither is a standout on novelty or patent potential (both MED), and both are present mainly on feasibility/budget/market-size merits rather than differentiation.

One domain (Cross-Cutting Enabling Tech) fully dropped out. This isn't a diversity gap in the usual sense — OC-30 (energy harvesting) and OC-31 (edge-AI research frontier) were eliminated for feasibility/skill-fit reasons specific to this team and timeline, not because the domain itself is weak (OC-31 in particular had the single highest research-gap score in the dataset). OC-30's harvesting concept remains available as a **feature**, not a **project** — see Section 2.

Diversification Recommendation

No forced diversification is recommended. The natural post-gate distribution is already reasonably balanced across the three largest domains (Healthcare, Industrial, Agriculture/Environment), and the "thin" domains are thin for defensible, quality-driven reasons rather than an artifact of the gate process. The one actionable recommendation is: **Phase 1C should not be shy about weighting Cybersecurity-Hardware (OC-21, OC-22) heavily despite its low count** — with only 2 candidates, it would take very little for this domain to be crowded out by sheer numerical minority in a naive vote-counting exercise, when in fact it represents the strongest alignment between team specialization, patent potential, and Phase 1B's own composite ranking.

5. Final Recommended Opportunity Clusters to Advance into Phase 1C

All 22 survivors are recommended to advance — the gate system already did the elimination work, and further cutting would risk discarding legitimate options before Phase 1C's idea-synthesis step has had a chance to combine them with specific technology building blocks. They are organized into two tiers to make the mandatory scope conditions impossible to miss.

Tier A — Advance Without Reservation (16 clusters)

No scope restrictions beyond ordinary engineering discipline. Ordered by Phase 1B composite score:

1. **OC-21** — Hardware Root-of-Trust & Secure Boot (8.3)
2. **OC-10** — Vibration-Based Predictive Maintenance (8.1)

3. **OC-15** — Precision Irrigation & Soil Sensing (7.9)
4. **OC-26** — Structural Health Monitoring (7.5)
5. **OC-01** — Offline RPM / Vital-Sign Monitoring (7.5)
6. **OC-05** — Cold-Chain / Pharma Logistics Monitoring (7.4)
7. **OC-12** — Industrial Process Quality Monitoring (7.4)
8. **OC-03** — Point-of-Care Diagnostics (7.3)
9. **OC-17** — Self-Calibrating Air/Water Quality (7.2)
10. **OC-13** — Worker Safety Monitoring (7.0)
11. **OC-06** — Hospital Asset Tracking + Security (6.9)
12. **OC-16** — Pest/Crop-Disease/Livestock Monitoring (6.9)
13. **OC-09** — Navigation/Obstacle Detection (VI) (6.7)
14. **OC-28** — Asset/Cargo/Tool Anti-Theft Tracking (6.6)
15. **OC-25** — Pothole/Road Condition Assessment (6.4)
16. **OC-20** — Wildlife & Biodiversity Monitoring (6.2)

Tier B — Advance With Mandatory Scope Narrowing (6 clusters)

High strategic value — three of these (OC-11, OC-08, OC-22) rank in Phase 1B's overall top 6 — but each carries a binding scope restriction from Section 3.2 that Phase 1C must respect, not revisit:

1. **OC-11** — Secure Offline IIoT/OT Anomaly Detection (8.2) — *single protocol, single anomaly demo*
2. **OC-08** — Assistive Communication & Mobility Devices (8.0) — *Braille/AAC/sign-language only, not EMG-prosthetic*
3. **OC-22** — Physical & Side-Channel Attack Defense (7.8) — *single narrow demo, confirm equipment access*
4. **OC-14** — Energy-Infrastructure Monitoring (7.7) — *solar/battery only, not live-line TDR*
5. **OC-02** — Non-Contact Fall Detection (7.6) — *IMU+acoustic primary, radar optional stretch only*
6. **OC-19** — Climate & Disaster Early-Warning Networks (7.3) — *single hazard, single node*

Board's Closing Observation

If Phase 1C needs to narrow further before generating concepts, the board's collective view (CTO, VC Partner, Cybersecurity Researcher, and Patent Examiner roles particularly) is that **OC-21, OC-11, OC-10, OC-08, OC-15, and OC-22** represent the six strongest bets by the combined weight of composite score, patent classification, and team-skill fit — four of

which sit in the team's stated cybersecurity specialization or its immediate adjacency (embedded security), which is a legitimate factor to weight even though it isn't one of the eight formal gates. This is offered as a steer, not a further elimination — Phase 1C should still work from the full 22-cluster survivor list.

Summary

- **32 clusters evaluated** against 8 gates (4 critical PASS/FAIL, 4 informational scored)
- **10 eliminated:** OC-04, OC-07, OC-18, OC-23, OC-24, OC-27, OC-29, OC-30, OC-31, OC-32
- **22 survive** (69% — above the anticipated 40–60%, explained by Phase 1B having already performed substantial quality filtering before this gate pass ever began): 16 clean, 6 conditional on explicit scope narrowing
- **Portfolio is reasonably diversified** across Healthcare, Industrial, and Agriculture/Environment (18–23% each); Cybersecurity-Hardware is numerically thin (9%) but disproportionately high-quality and should not be diluted in the name of raw domain balance
- **No new opportunities, products, architectures, or implementations were proposed** — this document stays strictly within Opportunity Portfolio Optimization, as instructed, and hands off 22 evidence-vetted clusters (2 tiers) to Phase 1C.